开源软件基础

基于 z3 的约束求解

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Outline

1. z3 简介

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概念

Z3 is a theorem prover from Microsoft Research. It is licensed under the MIT license[a].

[a] https://github.com/Z3Prover/z3

约束求解的历史

1. SAT: The satisfiability problem
2. SMT: Satisfiability modulo theories

约束求解示例

Consider the following puzzle. Spend exactly 100 dollars and buy exactly 100 animals. Dogs cost 15 dollars, cats cost 1 dollar, and mice cost 25 cents each. You have to buy at least one of each. How many of each should you buy?

约束求解示例

```
import z3
dog, cat, mouse = z3.Ints('dog cat mouse')
z3.solve(dog >= 1,    # at least one dog
      cat >= 1,       # at least one cat
      mouse >= 1,     # at least one mouse
      # we want to buy 100 animals
      dog + cat + mouse == 100,
      # We have 100 dollars (10000 cents):
      #   dogs cost 15 dollars (1500 cents),
      #   cats cost 1 dollar (100 cents), and
      #   mice cost 25 cents
      1500 * dog + 100 * cat + 25 * mouse == 10000)
```

程序生成/修复

```
from z3 import *
x = Int('x')
y = Int('y')
f = Function('f', IntSort(), IntSort())
s = Solver()
s.add(f(f(x)) == x, f(x) == y, x != y)
print s.check()
m = s.model()
print("f(f(x)) =", m.evaluate(f(f(x))))
print("f(x)    =", m.evaluate(f(x)))
```